

Statement of intent

Baha Khammari · PhD position, auction-based support schemes for renewable energy
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I want to write a doctorate on how the design of renewable support schemes changes what gets built. One question has brought me here, and I have met it twice already in my own work: the support an instrument appears to offer and the incentive it creates are different quantities, and the rules decide the distance between them. Energy market design is argued mostly in theory, while Europe has spent a decade running the experiments. Reading those experiments is work for an applied microeconomist, and it is the work I want to do.

My training is in economics at the University of Cologne, where the energy specialisation is two modules. *Growth, Energy, Climate Change* covers the physical and economic foundations of energy systems and European regulation and market intervention. *Energy Markets and Regulation*, with Jun.-Prof. Dr. Oliver Ruhnau, takes those into grids, interconnectors and the European electricity market, and I completed it with 1.3, where 1.0 is the maximum. Those modules are why an auction rule reads to me as a physical and regulatory constraint as much as an economic one.

My quantitative preparation is the strongest part of my record: 1.0 in mathematics for economists, 1.3 in econometrics, and 1.0, 1.0 and 1.3 in bachelor statistics, mathematics and applied econometrics. I have also tutored two school students through Abitur-level mathematics, which is the test of whether you know a method or only recognise it. My thesis, graded 1.0 and worth 30 of the degree's 120 ECTS, put it to work on a national scholarship programme in Brazil, on a panel I built from administrative records, five million linked employer-employee observations across 558 microregions, 2005 to 2019, in R through BigQuery. Callaway and Sant'Anna (2021) with doubly-robust inference gives a wage gain of 3.04 per cent in low-exposure regions, distinguishable from zero at the 10 per cent level, on a non-monotonic dose-response. What it settled is that dose, rather than treatment status, is where the answer sits, and that I can get there from raw records on my own.

The two directions I would take up are de-risking green investment and multi-technology auction design. My final seminar paper, a research design rather than a set of estimates, sits inside the first. It defines the effective EU ETS carbon cost as the allowance price attenuated by the sector-year free-allocation share, and shows the attenuation is not lump-sum, because activity thresholds and closure provisions tie allocation to continued production. A public intervention that weakens a cost incentive while the headline price holds is the mechanism your first direction names, and the same logic reaches guarantees, contracts for difference and public insurance. The second direction has the cleaner empirical footing. Since the 2016 state aid framework member states have arrived at technology-neutral, technology-specific and mixed designs at different dates. Germany's joint auction of April 2018 gave all 200 MW to solar at 46.7 euros per megawatt hour on average, some eight per cent above the solar-only auction two months before. Portfolio composition and rents are observable, which makes the question measurable.

Three people at CORE are why I want to do that there. Your work with Yueting Yu on discriminatory versus uniform-price auctions asks how short-run bidding maps into long-run investment, the rent question inside the second direction. Quentin L  t   works on market design, capacity investment and risk management, which is the first. ENERPOL, with Jean Hindriks and Fran  ois Maniquet, studies who carries the cost of the transition, which is what my thesis estimates. What I would bring to a group that argues these questions formally is their empirical counterpart, since European support-scheme reform is staggered adoption and I know how to read it. In ten years I want to be producing the evidence that European auction and support-scheme reform is judged on. Getting there needs graduate training in the theory this group works in and four years of supervised research on a question I already care about. This position is both.

M.Sc. Economics completed September 2026, available from October. French B1, German and Tunisian Arabic native, English IELTS Academic 8.0. EU citizen. Referees: Prof. Dr. Pia Pinger and Jun.-Prof. Dr. Oliver Ruhnau, contact details on the CV.

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